Federated Learning for Urban Noise Classification

Student Budget Edition

SoundTag Project — Zero-Cost Architecture

SoundTag ML Team — University Project

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1 Executive Summary

This document presents a **zero-cost federated learning architecture** for the SoundTag urban noise classification project, designed specifically for university students with:

- **Low-end Android phones** (budget devices, 2–4 GB RAM)
- **2 iPhones** (limited Android tooling)
- **No budget for GPUs or cloud servers**
- **Small team** (3–6 people max)
- **University timeline** (1 semester project, 12 weeks)

Key Takeaway

Core Philosophy: We use free GPU resources (Google Colab, Kaggle), lightweight models that run on potato phones, and a simplified federated learning approach that doesn't require complex infrastructure. The trade-off is longer training times and smaller models — but the result is still a working proof-of-concept suitable for a university thesis or project.

2 Existing Codebase Analysis

2.1 Current Architecture

The SoundTag project currently consists of two client applications:

Table 1: Current Application Components

Aspect	Android App (Kotlin)	Web App (Next.js/React)
Language	Kotlin	TypeScript/JavaScript
Framework	Jetpack Compose, Material 3	Next.js 15, React 19
Audio Capture	MediaRecorder (AAC, 44100Hz mono)	MediaRecorder API (WebM/Opus or MP4/AAC)
Min Duration	5 seconds	5 seconds
Max Duration	5 minutes	5 minutes
Metadata	GPS, timestamp, dB levels	GPS, timestamp
Storage	MediaStore API + SharedPreferences	IndexedDB + localStorage
Upload	Google Drive (OAuth)	Google Drive (OAuth)
Annotation	8 noise classes, severity, environment	Same as Android

2.2 Noise Label Categories

The system currently supports 8 annotated noise classes:

2.3 Data Collection Pipeline

Current flow:

1. User records audio (5s – 5min) via Android app or web browser
2. GPS coordinates captured automatically

Table 2: Noise Classification Categories

Class	Description / Examples
`traffic`	Cars, buses, rickshaws, general road noise
`horn`	Vehicle honking (specifically honking sounds)
`construction`	Drilling, hammering, building activity
`industrial`	Factory hum, generators, machinery
`crowd`	Market noise, protests, large gatherings
`nature`	Rain, wind, birds, natural sounds
`silence`	Quiet baseline, non-noise reference
`mixed`	Multiple simultaneous noise sources

3. User annotates: noise label, severity (low/medium/high), environment (indoor/outdoor), location context
4. Audio (.m4a or .webm) + JSON metadata sidecar saved locally
5. Optional upload to Google Drive: `SoundTag/{annotator_id}/`
6. Dashboard shows recording history with dataset balance warnings

3 Technology Stack: Librosa for Audio Feature Extraction

3.1 Why Librosa

Librosa is the industry-standard Python library for audio analysis. For this project:

- **Mel Spectrogram**: Converts raw audio to time-frequency representation matching human auditory perception
- **MFCCs**: Compact feature representation widely used in audio classification
- **Feature Utilities**: Zero-crossing rate, spectral centroid, spectral rolloff, chroma features
- **Free and Open Source**: No cost, well-documented

3.2 Preprocessing Pipeline

```python
import librosa
import numpy as np

def extract_features(audio_path, sr=22050, n_mfcc=13):
    # Load audio, resample to 22050 Hz (standard for ML)
    y, sr = librosa.load(audio_path, sr=sr)

    # Core features
    mel_spec = librosa.feature.melspectrogram(
        y=y, sr=sr, n_mels=64, fmax=8000
    )
    mel_spec_db = librosa.power_to_db(mel_spec, ref=np.max)

    mfccs = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=n_mfcc)
    zcr = librosa.feature.zero_crossing_rate(y)

    # Stack: shape (78, T)     smaller for student-scale models
    features = np.vstack([mel_spec_db, mfccs, zcr])
    return features, sr

# Output shape: (78, T) where T depends on audio duration
```

4 Edge Device Model Recommendations (Potato Phones)

4.1 Reality Check: Constraints

Table 3: Realistic Edge Device Constraints

Constraint	Reality
Devices	Low-end Android (2–4 GB RAM), 2 iPhones
Memory Budget	$\leq$ 20 MB RAM for ML
Model Size	$\leq$ 5 MB (downloadable over slow WiFi)
Inference Time	$\leq$ 2 seconds per 5s clip (acceptable)
CPU	Snapdragon 4xx, MediaTek Helio G series
Training on Device	**NOT FEASIBLE** on potato phones

Key Takeaway

Key Decision: On-device training is NOT practical on low-end phones. Instead, we use a **simplified federated learning** approach where:

1. Edge devices only do **inference** (prediction)
2. Training happens on **free cloud GPUs** (Google Colab / Kaggle)
3. Model updates are pushed back to devices
4. This is still "federated" because data stays on devices, only labels + features go to the server

4.2 Recommended Edge Models (Ranked by Phone Friendliness)

4.2.1 Option 1: Tiny CNN — Custom Lightweight Model (Recommended)

- **Architecture**: 3 convolutional layers + 2 dense layers
- **Parameters**: ~200K – 500K (tiny!)
- **Model Size**: ~1 – 2 MB (FP32), ~0.5 – 1 MB (INT8 quantized)
- **Inference Speed**: 500 ms – 2 seconds on low-end Android
- **Framework**: TFLite (Android), TFLite.js / ONNX.js (Web)
- **Quantization**: INT8 post-training quantization for 4× size reduction

Why this works: Audio classification doesn't need a huge model. A small CNN trained on mel spectrograms can achieve 70–85% accuracy on 8 noise classes with only 200K parameters.

4.2.2 Option 2: YAMNet Embedding Extractor

- **Architecture**: Pre-trained YAMNet (MobileNetV1 on AudioSet) used as feature extractor
- **Approach**: Run YAMNet on device to extract 1024-D embedding, send embedding to server for classification
- **Advantage**: YAMNet is already trained on 527 audio classes — good feature extractor
- **Disadvantage**: YAMNet itself is 4 MB+; may be slow on potato phones
- **Use Case**: If phones can handle YAMNet inference, this is a strong option

4.2.3 Option 3: No On-Device Model — Server-Only Inference

- **Architecture**: Audio recorded on phone → uploaded to server → server classifies → returns prediction
- **Advantage**: Zero ML on device, works on any phone
- **Disadvantage**: Requires internet, slower (upload + inference time)
- **Use Case**: Fallback for iPhones and very low-end Android devices

4.3 iPhone Consideration

- iPhones cannot run TFLite easily (Apple prefers Core ML)
- **Solution**: Use the **web app** for iPhones — TF.js runs in Safari/Chrome
- Or: Use server-only inference (Option 3) for iPhone users
- Since you only have 2 iPhones, this is a minor concern

4.4 Final Recommendation for Edge Model

Recommended: Custom Tiny CNN (3 conv layers, ∼200K params) exported as TFLite.
Fallback: Server-only inference for devices that can't run the model locally.
No on-device training. Training happens on free cloud GPUs.
Reasoning:

- Small enough for potato phones (1–2 MB model, ≤2s inference)
- Simple architecture — easy to implement for students
- Training centralized on free GPU (Colab/Kaggle)
- Still federated: data stays on devices, only aggregated labels/features go to server

5 Central Server — Zero-Cost Architecture

5.1 Free GPU Options

Table 4: Free GPU Resources for Students

Platform	GPU	Limits
Google Colab (Free)	T4 (16GB) or K80 (12GB)	12-hour sessions, may disconnect, queue times during peak hours
Kaggle Notebooks	T4 × 2 (16GB) or P100 (16GB)	30 hours/week, 9-hour session limit
Google Colab Pro ($10/mo)	V100 (16GB), sometimes A100	Priority access, longer sessions
University Lab GPU	Varies	Check if your CS department has a GPU server

Key Takeaway

Recommended: Use **Google Colab Free** for training. It provides a T4 GPU with 16GB VRAM for free. Sessions last up to 12 hours, which is enough for training a small model (training takes 30 minutes – 2 hours). Use Kaggle Notebooks as backup.

5.2 Central Model Architecture (Student-Scale)

- **Architecture**: Tiny CNN — 3 conv layers + global average pooling + dense classifier
- **Parameters**: ∼500K – 1M (much smaller than industry-scale)
- **Framework**: PyTorch (Colab-friendly) or TensorFlow (easier TFLite export)
- **Input**: Mel spectrogram (64 frequency bins × time frames), resized to 64 × 64
- **Output**: 8-class softmax

5.3 Server Infrastructure — Simplified

Since we can't afford a always-on server, we use a **pull-based** approach:

1. **Training**: Done on Google Colab when needed (manual trigger)
2. **Model Storage**: Trained model saved to Google Drive (free 15 GB)
3. **Model Distribution**: Clients download model files from Google Drive link
4. **Data Aggregation**: Labeled data collected from Google Drive uploads (existing SoundTag pipeline)
5. **No always-on server needed** — everything runs on-demand

```
+-------------------------------------------------------------+
|                ZERO-COST FEDERATED LEARNING FLOW             |
+-------------------------------------------------------------+

Step 1: Data Collection (Ongoing)

 User records audio on phone → annotates label
 Audio + JSON metadata uploaded to Google Drive
 (Existing SoundTag pipeline - no changes needed)



Step 2: Training (Weekly/Bi-weekly, on Google Colab)

 Open Colab notebook → download all data from Drive
 Extract mel spectrograms with librosa
 Train Tiny CNN model on T4 GPU (free)
 Training takes 30 min -- 2 hours
 Export trained model to TFLite format
 Upload model to Google Drive



Step 3: Model Distribution

 Share updated TFLite model via Google Drive link
 Android app: download TFLite file, replace old model
 Web app: load new TF.js model from Drive/static folder
 iPhones: use web app or server inference fallback



Step 4: On-Device Inference (Automatic Classification)
```

```
User records audio → run TFLite model → get prediction
Show predicted label + confidence to user
User confirms or corrects
Corrections become new training data for next round
```

6 Simplified Federated Learning Approach

6.1 Why Not "True" Federated Learning?

Table 5: True FL vs. Simplified FL

Aspect	True Federated Learning	Simplified FL (Our Approach)
On-device training	Yes (FedAvg, weight updates)	**No** (too heavy for potato phones)
Data stays on device	Yes	Yes
What goes to server	Model weights (encrypted)	Labels + audio files (via Drive)
Server infrastructure	Always-on FL server	**On-demand Colab notebook**
Cost	$1,000+/month	**$0**
Complexity	High (Flower, TFF, weight serialization)	**Low** (train → export → distribute)
Suitable for students	No	**Yes**

Key Takeaway

This is still federated learning in spirit: The model improves by learning from distributed data without centralizing raw audio. The key FL principles are preserved:

- Data decentralization (audio stays on devices)
- Collaborative model improvement (all users' data contributes)
- Privacy-preserving (no raw audio sharing beyond what users opt to upload)

For a university project, this approach is practical, achievable, and still demonstrates the core FL concept.

6.2 Alternative: Feature-Based FL (If You Want More FL Flavor)

If you want to make it more "federated" without on-device training:

1. On device: Extract mel spectrogram features from audio (no label needed)
2. Send **features only** (not raw audio) to a central Colab notebook
3. Central server trains on aggregated features + labels from users who annotated
4. This way, even unannotated recordings contribute (semi-supervised learning)

This approach reduces bandwidth (features are smaller than audio files) and allows semi-supervised learning.

7 Dataset Size Requirements (Student-Scale)

7.1 Realistic Targets for a Small Team

Table 6: Dataset Size Recommendations (Student Project)

Phase	Samples per Class	Total Samples	Total Hours (est.)
Proof of Concept	50	400 (8 classes)	∼30 minutes
Minimum Viable	200	1,600 (8 classes)	∼2.2 hours
Good for Thesis	500	4,000 (8 classes)	∼5.5 hours
Ambitious	1,000	8,000 (8 classes)	∼11 hours

With 4–6 annotators collecting 5 recordings/day each:

- **Proof of Concept**: 1–2 days of collection
- **Minimum Viable**: 1 week
- **Good for Thesis**: 2–3 weeks
- **Ambitious**: 4–6 weeks

7.2 Data Collection Strategy for Students

1. **Week 1**: Each team member collects 50 samples (traffic-heavy, your campus/commute)
2. **Week 2**: Expand to different locations (markets, residential areas, industrial zones)
3. **Week 3**: Focus on underrepresented classes (nature, silence — these are harder to find)
4. **Week 4+**: Continuous collection, start training models

7.3 Class Balance Considerations

- **Easy to collect**: Traffic, horn, crowd (common in urban areas like Dhaka)
- **Hard to collect**: Nature, silence, industrial (need specific locations)
- **Strategy**: If a class has <50 samples, consider merging it:
  - `nature` + `silence` → `non-noise`
  - `industrial` → merge into `construction`
- This reduces from 8 classes to 5–6 classes, making training easier with limited data

8 Complete Implementation Plan (Student-Friendly, 12 Weeks)

8.1 Phase 1: Data Collection (Week 1–3)

1. **Use existing SoundTag apps as-is** — no ML changes yet
2. Each team member records 50–100 annotated clips per week
3. Target: 1,600+ samples (200 per class) by end of Week 3
4. All uploads go to Google Drive (existing pipeline)
5. **Deliverable**: Dataset of 1,600+ labeled audio clips in Google Drive

8.2 Phase 2: Central Training on Colab (Week 4–5)

1. **Set up Google Colab notebook**
   - Download all audio + JSON files from Drive
   - Extract mel spectrograms using librosa

- Split into train/val/test (70/15/15)

2. **Build and train Tiny CNN model**
   - PyTorch or TensorFlow model (code below)
   - Train on Colab's free T4 GPU
   - Target: ≥70% validation accuracy
   - Training time: 30 minutes – 2 hours
3. **Export model**
   - TensorFlow → TFLite (INT8 quantized)
   - Save TFLite model to Google Drive
   - Upload to GitHub repo as release artifact
4. **Deliverable**: Trained TFLite model in Google Drive, Colab notebook with full training pipeline

8.3 Phase 3: Android App ML Integration (Week 6–8)

1. **Add TFLite dependency to Android app**

```
// app/build.gradle.kts
dependencies {
    implementation("org.tensorflow:tensorflow-lite:2.14.0")
    implementation("org.tensorflow:tensorflow-lite-support:0.4.4")
}
```

2. **Download model from Google Drive (or bundle in app)**
   - Option A: Bundle TFLite model in app assets (simplest, 1–2 MB)
   - Option B: Download model from Drive link on first launch
   - Option C: Check for model updates periodically
3. **Implement mel spectrogram extraction on Android**
   - Use `android.media.AudioRecord` to get raw PCM
   - Implement mel spectrogram in Kotlin (or use a Java port of librosa's mel filterbank)
   - Alternative: Use **TensorFlow Lite Audio Task API** which handles feature extraction internally
4. **Implement inference pipeline**
   (a) User records audio
   (b) Extract mel spectrogram
   (c) Run TFLite model → get class probabilities
   (d) Show predicted label + confidence
   (e) User confirms or corrects
5. **Deliverable**: Android app with on-device noise classification

8.4 Phase 4: Web App ML Integration (Week 7–8, parallel with Phase 3)

1. **Add TensorFlow.js to web app**

```
// package.json
"@tensorflow/tfjs": "^4.11.0",
"@tensorflow-models/audio-commands": "^0.2.6" // optional
```

2. **Convert model to TF.js format**

```
# In Colab, after training:
tensorflowjs_converter --input_format=tf_saved_model \
    --output_format=tfjs_graph_model \
    ./saved_model/ ./tfjs_model/
```

3. **Implement mel spectrogram in browser**
   - Use **meyda.js** library: `npm install meyda`
   - Or use Web Audio API `AnalyserNode` for FFT-based spectrogram
4. **Implement inference pipeline**
   (a) User records audio via existing recorder.js
   (b) Extract mel spectrogram using meyda.js
   (c) Load TF.js model, run inference
   (d) Show predicted label + confidence
   (e) User confirms or corrects
5. **Deliverable**: Web app with browser-based noise classification

8.5 Phase 5: Auto-Classification + Active Learning Loop (Week 9–10)

1. **Confidence-based prediction**
   - Confidence $\geq$ 80%: auto-accept label (no user input needed)
   - Confidence 60–80%: show prediction, allow quick correction
   - Confidence $<$ 60%: fall back to full manual annotation
2. **Active learning loop**
   - Every correction is a new training sample
   - Collect corrections locally
   - Weekly: upload all new labeled data to Drive
   - Retrain model on Colab with expanded dataset new model version, distribute to clients
3. **Track improvement**
   - Log prediction accuracy (how often was model correct?)
   - Log correction rate (how often did users override?)
   - Plot accuracy vs. rounds — should improve over time

8.6 Phase 6: Thesis Writing and Demo Preparation (Week 11–12)

1. **Collect results**: Model accuracy over rounds, confusion matrix, per-class F1 scores
2. **Write thesis**: Architecture, methodology, results, discussion
3. **Prepare demo**: Live demo of recording $\rightarrow$ auto-classification $\rightarrow$ correction
4. **Prepare presentation**: Show model improvement over time, dataset statistics
5. **Deliverable**: Complete thesis, working demo, presentation slides

9 Code Samples (Student-Ready)

9.1 Tiny CNN Model (TensorFlow, Colab)

```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers

def build_tiny_cnn(input_shape=(64, 64, 1), num_classes=8):
    """Tiny CNN for mel spectrogram classification.
```

```python

    Input: Mel spectrogram resized to 64x64 (grayscale)
    Output: 8-class softmax
    Total params: ~300K (small enough for potato phones)
    """
    model = keras.Sequential([
        # Input: (64, 64, 1)
        layers.Conv2D(16, 3, activation='relu', padding='same'),
        layers.MaxPooling2D(2),  # (32, 32, 16)
        layers.Conv2D(32, 3, activation='relu', padding='same'),
        layers.MaxPooling2D(2),  # (16, 16, 32)
        layers.Conv2D(64, 3, activation='relu', padding='same'),
        layers.GlobalAveragePooling2D(),  # (64,)
        layers.Dense(32, activation='relu'),
        layers.Dropout(0.5),
        layers.Dense(num_classes, activation='softmax')
    ])
    return model

model = build_tiny_cnn()
model.summary()
# Total params: ~280,000

# Compile
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=0.001),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Train (on Colab T4 GPU)
history = model.fit(
    train_dataset,  # batched (mel_spec, label) pairs
    validation_data=val_dataset,
    epochs=30,
    callbacks=[
        keras.callbacks.EarlyStopping(
            monitor='val_loss', patience=5, restore_best_weights=True
        ),
        keras.callbacks.ReduceLROnPlateau(
            monitor='val_loss', factor=0.5, patience=3
        )
    ]
)

# Export to TFLite (INT8 quantized for smallest size)
converter = tf.lite.TFLiteConverter.from_keras_model(model)
converter.optimizations = [tf.lite.Optimize.DEFAULT]
tflite_model = converter.convert()

with open('soundtag_model.tflite', 'wb') as f:
    f.write(tflite_model)

# Expected model size: ~300 KB -- 1 MB (INT8 quantized)
```

9.2 Audio Preprocessing (Librosa, Colab)

```python
import librosa
import numpy as np
import os
import json

```

```python
def prepare_dataset(audio_dir, target_size=(64, 64)):
    """Load all audio files and JSON metadata, prepare dataset.

    Returns:
        X: numpy array of mel spectrograms (N, 64, 64, 1)
        y: numpy array of labels (N,)
        label_map: dict mapping class names to integers
    """
    classes = ['traffic', 'horn', 'construction', 'industrial',
               'crowd', 'nature', 'silence', 'mixed']
    label_map = {c: i for i, c in enumerate(classes)}

    X, y = [], []

    for filename in os.listdir(audio_dir):
        if not filename.endswith(('.m4a', '.webm', '.wav')):
            continue

        # Load audio
        audio_path = os.path.join(audio_dir, filename)
        audio_json_path = audio_path.rsplit('.', 1)[0] + '.json'

        if not os.path.exists(audio_json_path):
            continue

        # Load label from JSON
        with open(audio_json_path) as f:
            metadata = json.load(f)
        label = metadata.get('label', 'mixed')

        if label not in label_map:
            continue

        # Extract mel spectrogram
        audio_data, sr = librosa.load(audio_path, sr=22050, mono=True)
        mel_spec = librosa.feature.melspectrogram(
            y=audio_data, sr=sr, n_mels=64, fmax=8000
        )
        mel_spec_db = librosa.power_to_db(mel_spec, ref=np.max)

        # Resize to fixed size (64, 64)
        import cv2
        mel_resized = cv2.resize(mel_spec_db, target_size, interpolation=cv2.INTER_LINEAR)

        # Normalize to [0, 1]
        mel_resized = (mel_resized - mel_resized.min()) / (mel_resized.max() - mel_resized.min() + 1e-8)

        X.append(mel_resized)
        y.append(label_map[label])

    X = np.array(X)[..., np.newaxis]  # (N, 64, 64, 1)
    y = np.array(y)

    print(f"Dataset: {len(X)} samples, {len(classes)} classes")
    print(f"Class distribution: {np.bincount(y)}")

    return X, y, label_map
```

9.3 Android TFLite Inference (Kotlin)

```
import org.tensorflow.lite.Interpreter
import java.nio.ByteBuffer
import java.nio.ByteOrder

class SoundTagClassifier(assetManager: AssetManager, modelPath: String) {

    private val interpreter: Interpreter
    private val inputSize = 64  // 64x64 mel spectrogram
    private val numClasses = 8
    private val labelNames = listOf(
        "traffic", "horn", "construction", "industrial",
        "crowd", "nature", "silence", "mixed"
    )

    init {
        val modelFile = assetManager.open(modelPath)
        val modelBytes = modelFile.readBytes()
        val modelBuffer = ByteBuffer.allocateDirect(modelBytes.size)
        modelBuffer.order(ByteOrder.nativeOrder())
        modelBuffer.put(modelBytes)
        interpreter = Interpreter(modelBuffer)
    }

    fun classify(melSpectrogram: FloatArray): PredictionResult {
        // Input: FloatArray of size 64*64 = 4096
        // Output: FloatArray of size 8 (class probabilities)
        val inputBuffer = ByteBuffer.allocateDirect(4096 * 4)
        inputBuffer.order(ByteOrder.nativeOrder())
        for (value in melSpectrogram) {
            inputBuffer.putFloat(value)
        }
        inputBuffer.rewind()

        val outputBuffer = Array(1) { FloatArray(numClasses) }
        interpreter.run(inputBuffer, outputBuffer)

        val probabilities = outputBuffer[0]
        val maxIndex = probabilities.indices.maxByOrNull { probabilities[it] } ?: 0

        return PredictionResult(
            label = labelNames[maxIndex],
            confidence = probabilities[maxIndex],
            allProbabilities = labelNames.zip(probabilities.toList()).toMap()
        )
    }

    data class PredictionResult(
        val label: String,
        val confidence: Float,
        val allProbabilities: Map<String, Float>
    )
}

// Usage in your Recording Activity:
// val classifier = SoundTagClassifier(assets, "soundtag_model.tflite")
// val result = classifier.classify(melSpectrogramFloatArray)
// showPrediction(result.label, result.confidence)
```

9.4 Web App TF.js Inference (JavaScript)

```
import * as tf from '@tensorflow/tfjs';
```

```javascript
import Meyda from 'meyda';

class SoundTagClassifier {
  constructor() {
    this.model = null;
    this.classes = [
      'traffic', 'horn', 'construction', 'industrial',
      'crowd', 'nature', 'silence', 'mixed'
    ];
  }

  async loadModel(modelUrl) {
    this.model = await tf.loadGraphModel(modelUrl);
  }

  extractMelSpectrogram(audioBuffer, nMels = 64, targetSize = 64) {
    const audioData = audioBuffer.getChannelData(0);
    const features = Meyda.extract(
      ['melSpectrogram'],
      {
        ...audioBuffer,
        sampleRate: audioBuffer.sampleRate,
        bufferSize: 1024,
        hopSize: 512,
      }
    );

    // Meyda returns melSpectrogram as array of arrays
    // Resize to fixed targetSize x targetSize
    const melSpec = features.melSpectrogram;
    const resized = this.resize2D(melSpec, targetSize, targetSize);

    // Normalize to [0, 1]
    const min = Math.min(...resized.flat());
    const max = Math.max(...resized.flat());
    const normalized = resized.map(row =>
      row.map(v => (v - min) / (max - min + 1e-8))
    );

    return tf.tensor(normalized).reshape([1, targetSize, targetSize, 1]);
  }

  resize2D(matrix, newRows, newCols) {
    const oldRows = matrix.length;
    const oldCols = matrix[0].length;
    const result = [];
    for (let i = 0; i < newRows; i++) {
      const row = [];
      for (let j = 0; j < newCols; j++) {
        const oldI = Math.floor((i / newRows) * oldRows);
        const oldJ = Math.floor((j / newCols) * oldCols);
        row.push(matrix[oldI][oldJ]);
      }
      result.push(row);
    }
    return result;
  }

  async predict(audioBuffer) {
    const inputTensor = this.extractMelSpectrogram(audioBuffer);
    const prediction = this.model.predict(inputTensor);
    const probs = await prediction.data();
    inputTensor.dispose();
```

```
    prediction.dispose();

    const maxIndex = probs.indexOf(Math.max(...probs));
    return {
      label: this.classes[maxIndex],
      confidence: probs[maxIndex],
      allProbabilities: Object.fromEntries(
        this.classes.map((c, i) => [c, probs[i]])
      )
    };
  }
}

// Usage in your web app:
// const classifier = new SoundTagClassifier();
// await classifier.loadModel('/models/soundtag_model.json');
// const audioBuffer = await audioContext.decodeAudioData(audioBlob);
// const result = await classifier.predict(audioBuffer);
// console.log(`Predicted: ${result.label} (${(result.confidence * 100).toFixed
    (1)}%)`);
```

9.5 Colab Training Notebook Structure

```
SoundTag_Training_Notebook.ipynb
 Cell 1: Mount Google Drive
   from google.colab import drive
   drive.mount('/content/drive')

 Cell 2: Install dependencies
   !pip install librosa tensorflow opencv-python-headless

 Cell 3: Load data from Drive
   X, y, label_map = prepare_dataset('/content/drive/MyDrive/SoundTag/data/')

 Cell 4: Split train/val/test
   from sklearn.model_selection import train_test_split
   X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3)
   X_train, X_val, y_train, y_val = train_test_split(X_train, y_train, test_size=0.2)

 Cell 5: Build model
   model = build_tiny_cnn()

 Cell 6: Train
   history = model.fit(X_train, y_train, validation_data=(X_val, y_val), epochs=30)

 Cell 7: Evaluate
   model.evaluate(X_test, y_test)
   Plot confusion matrix, per-class accuracy

 Cell 8: Export to TFLite
   converter = tf.lite.TFLiteConverter.from_keras_model(model)
   converter.optimizations = [tf.lite.Optimize.DEFAULT]
   tflite_model = converter.convert()
   Save to Drive: /content/drive/MyDrive/SoundTag/models/soundtag_v1.tflite
```

```
Cell 9: Download link for Android/Web app
   # Generate shareable Drive link for the model
```

10 Cost Estimation (Zero Budget)

Table 7: Cost Breakdown — Student Edition

Item	Cost	Notes
Training GPU	$0	Google Colab Free (T4, 12-hour sessions)
Backup GPU	$0	Kaggle Notebooks (30 hrs/week free)
Storage	$0	Google Drive (15 GB free)
Model Distribution	$0	Google Drive links / GitHub Releases
API Server	$0	Not needed (pull-based model updates)
Colab Pro (optional)	$10/mo	If free tier has queue issues
Total (Monthly)	**$0 – $10**	Basically free!

11 Risk Mitigation (Student Context)

Table 8: Student-Specific Risks and Mitigation

Risk	Impact	Mitigation
Colab disconnects mid-training	High: Lose training progress	Save checkpoints every epoch; use ModelCheckpoint callback to Drive
Too few samples per class	High: Model underperforms	Merge rare classes; collect more data; use data augmentation
Potato phones can't run TFLite	Medium: Fallback needed	Bundle smallest model; use server inference fallback
iPhone compatibility issues	Low: Only 2 iPhones	Use web app (TF.js in Safari); server inference fallback
Team members lose motivation	Medium: Slow progress	Set weekly targets; celebrate milestones; divide work clearly
Model accuracy too low for thesis	High: Not defendable	Reduce classes (8→5); focus on top 3 classes; show improvement over rounds

12 Timeline — 12 Week Plan

Table 9: Week-by-Week Timeline

Week	Phase	Deliverables
1–3	Data Collection	1,600+ labeled audio clips in Drive
4–5	Central Training	Colab notebook, trained TFLite model, ≥70% val accuracy
6–8	Client ML Integration	Android + Web apps with on-device classification
9–10	Active Learning Loop	Auto-classification, correction tracking, retraining cycle
11–12	Thesis + Demo	Complete thesis, working demo, presentation slides

13 Conclusion

Key Takeaway

This plan is designed for your reality:

- **No budget** → Free Colab GPUs, free Drive storage, no always-on server
- **Potato phones** → Tiny CNN model (∼300K params, 1–2 MB), or server inference fallback
- **Small team** → 1,600–4,000 samples total (not 40,000), 5–6 classes instead of 8 if needed
- **12-week timeline** → Realistic milestones, proof-of-concept by Week 5, thesis-ready by Week 12

13.1 What You'll Have at the End

1. A working Android app that auto-classifies noise on-device
2. A working web app with browser-based classification
3. A trained model that improves over time with more data
4. A Colab notebook for retraining (reproducible pipeline)
5. A thesis documenting the federated learning approach and results

13.2 What Makes This Impressive for a University Project

- **Real-world problem**: Urban noise pollution is a genuine issue
- **Federated learning concept**: Data stays on devices, model improves collaboratively
- **End-to-end pipeline**: Data collection → training → deployment → improvement loop
- **Working demo**: Live classification on real phones
- **Zero budget**: Shows resourcefulness and engineering skill

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